Lemma 4.5 (Accumulated drift and diminishing returns)**.** *Assume mean-consistency and $\delta_{i,t} \geq \mu$ for $t = 0, \ldots, T-1$. Let $S_{i,0} := \|\alpha_{i,0}\|_1$, then*

$$\begin{aligned}\mathbb{E}[p_{i,T}] &\geq p_{i,0} + \mu \sum_{t=1}^{T} \frac{1}{S_{i,0} + tC_i} \\ &\geq p_{i,0} + \frac{\mu}{C_i} \log\left(\frac{S_{i,0} + (T+1)C_i}{S_{i,0} + C_i}\right).\end{aligned} \tag{12}$$

Proposition 4.6 (Training increases critique advantage)**.** *Suppose debate training yields $\delta_{i,t} \geq \mu > 0$ on the evaluation-time debate distribution for early rounds. Then Theorem 4.4 and Lemma 4.5 imply $\mathbb{E}[p_{i,t}]$ increases with t.*

Lemma 4.7 (Correlation shrinks the effective ensemble size)**.** *Consider a single debate round with N agents producing answers $Y_1, \ldots, Y_N \in \{1, \ldots, K\}$. Assume each Y_n has the same marginal distribution $p \in \Delta^K$ with gap $\Delta := p_1 - p_2 > 0$ ((Choi et al., 2025) Theorem 1). Let $\hat{p}$ be the empirical distribution and $y_{\mathrm{mv}} = \arg\max_k \hat{p}_k$ be the majority vote.*

Let the correlation parameter be

$$\rho := \max_{k \in [K]} \max_{a \neq b} \frac{\mathrm{Cov}(\mathbb{1}\{Y_a = k\}, \mathbb{1}\{Y_b = k\})}{p_k(1 - p_k)} \in [0, 1].$$

with the assumption that $p_k \in (0, 1)$ for all k. Then the plurality-vote error probability admits the bound

$$\Pr(y_{\mathrm{mv}} \neq 1) \leq \frac{K(1 + (N-1)\rho)}{N\,\Delta^2} = \frac{K}{N_{\mathrm{eff}}\Delta^2},$$
$$N_{\mathrm{eff}} := \frac{N}{1 + (N-1)\rho}.$$

Role of majority vs. private critique. Our analysis disentangles the contributions of *majority voting* and *private critique* in multi-round debate. When there is no training signal from critique (*i.e.,* $\delta_{i,t} = 0$), Theorem 4.4 reduces to a neutral (martingale) belief evolution, and the end-to-end accuracy is governed purely by vote amplification—recovering the standard majority-voting perspective (*e.g.,* (Choi et al., 2025)). SDRL helps precisely by increasing the critique advantage δ, which induces positive drift in each agent's correctness; however, Lemma 4.5 shows that even under sustained advantage ($\delta_{i,t} \geq \mu$), the improvement accumulates only logarithmically in the number of rounds and yields diminishing per-round gains. At the same time, as rounds progress agents condition on increasingly similar contexts, raising answer correlation and effectively reducing the ensemble size (Lemma 4.7), which weakens majority-vote amplification. Together, these two effects explain a common empirical pattern: performance improves in early rounds (positive critique drift with low correlation) but peaks quickly and can decline later as marginal drift shrinks and homogeneity erodes the benefits of voting.

5. Methodology

We propose Self-Debate Reinforcement Learning (SDRL), an online reinforcement learning framework that trains a *single* policy to improve both (i) single-agent reasoning and (ii) *private critique* required by Multi-Agent Debate (MAD). Section 5.1 introduces how SDRL uses verifiable rewards to train the model for single-agent reasoning. Section 5.2 introduces how SDRL equips a single model with the private critique ability to discriminate between different opinions during debate interactions.

5.1. Single-agent reasoning

To improve the model's reasoning ability in the single-agent setting, SDRL follows the GRPO framework (Shao et al., 2024) and optimizes the policy using verifiable rewards. For each prompt q, SDRL samples n independent responses $\{o_i\}_{i=1}^n \sim \pi_\theta(\cdot \mid q)$ from the current policy and assigns a sparse outcome reward $r_j \in \{+1, -1\}$ based on the correctness of the final answer. We then compute advantages for each response using Eq. 6. To prepare candidate responses for debate training, we follow DAPO (Yu et al., 2025) to oversample prompts and filter out those whose responses yield zero advantages. This ensures that each prompt has responses corresponding to different final answers, capturing diverse opinions and reasoning trajectories.

5.2. Debate training

The goal of SDRL's debate component is to explicitly train the policy to acquire private critique ability to discriminate among diverse reasoning trajectories and leverage them to produce a final answer. Given the initial rollout set for a prompt q, we construct a debate pair by selecting two responses from the candidate response pool, which is formed from the first-round responses generated for single-agent reasoning. Selecting two responses to form a debate pair is a computationally efficient design choice that induces the fundamental private critique behavior and generalizes to settings with more candidate responses.

Debate pair construction. Let $\mathcal{O}(q) = \{o_i\}_{i=1}^n$ denote the initial rollouts, which we use as the candidate pool, and let $a(o_i)$ be the extracted final answer of rollout o_i. SDRL constructs a debate pair $\mathcal{P}(o^1, o^2)$ by selecting two candidates from $\mathcal{O}(q)$ using one of two pairing rules:

1. *Random pairing*. It samples two rollouts uniformly from $\mathcal{O}(q)$. This induces a broad distribution of pair diversity. Some pairs exhibit genuine disagreement, while others share the same final answer but differ in their reasoning trajectories, which can provide useful training signals for confirmation rather than unconditional revision.

2. *Frequency-based pairing*. It first identifies the most common answer a_1 and the second most common answer a_2 in $\{a(o_i)\}_{i=1}^n$. This rule more consistently exposes the model to the dominant competing beliefs expressed by its current policy, yielding sharper disagreement instances and therefore requiring stronger private critique ability.

We evaluate both constructions and use the same downstream prompting and optimization for either choice.

Debate prompt formulation. Given a selected pair $\mathcal{P}(o^1, o^2)$, we build a debate prompt q_d by serializing the two candidate responses into a two-turn conversation, and then prompt the model to deliberate over both responses and produce a final answer. The detailed conversation format and prompt template are shown in the Appendix. For *frequency-based pairing*, we randomize the order of a_1 and a_2 to prevent positional heuristics. Since the most common answer a_1 typically has a higher probability of being correct, always placing a_1 as the first response in q_d could encourage the model to select the first-position answer rather than perform critical analysis of the underlying reasoning trajectories.

Second-turn rollouts and joint optimization. Conditioned on q_d, SDRL samples n_d second-turn responses $\{o_j^d\}_{j=1}^{n_d} \sim \pi_\theta(\cdot \mid q_d)$ and score them using the same verifiable outcome reward. SDRL computes advantages using Equation 6 across responses that share the same debate prompt q_d. Finally, SDRL merges the initial rollouts from q and the debate rollouts from q_d into a single training batch. For efficiency, we filter out debate responses o_j^d with zero advantage. Overall, the SDRL framework is summarized in Algorithm 1.

Algorithm 1 Self-Debate Reinforcement Learning (SDRL)

Input: Policy π_θ, prompt set $\mathcal{D} = \{q_i\}_{i=1}^N$, initial rollouts per prompt n, debate rollouts per prompt n_d
Output: Updated policy $\pi_{\theta_{\text{updated}}}$

for $t = 1, 2, \ldots, K$ **do**
  Sample a mini-batch $\mathbf{q} \subseteq \mathcal{D}$ and initialize batch $\mathbf{B} \leftarrow \emptyset$
  for each $q \in \mathbf{q}$ **do**
    Sample initial rollouts $\{o_i\}_{i=1}^n \sim \pi_\theta(\cdot \mid q)$
    Compute rewards and advantages $\{A_i\}_{i=1}^n$ for $\{o_i\}_{i=1}^n$
    $\mathbf{B} \leftarrow \mathbf{B} \cup \{(q, \{o_i\}_{i=1}^n, \{A_i\}_{i=1}^n)\}$
    Select a debate pair $\mathcal{P}(o^1, o^2)$ from $\{o_i\}_{i=1}^n$ and build debate prompt q_d
    Sample debate rollouts $\{o_j^d\}_{j=1}^{n_d} \sim \pi_\theta(\cdot \mid q_d)$
    Compute rewards and advantages $\{A_j^d\}_{j=1}^{n_d}$ for $\{o_j^d\}_{j=1}^{n_d}$
    $\mathbf{B} \leftarrow \mathbf{B} \cup \{(q_d, \{o_j^d\}_{j=1}^{n_d}, \{A_j^d\}_{j=1}^{n_d})\}$
  end for
  Update π_θ using the RL optimizer on $\mathbf{B}$
end for
return $\pi_{\theta_{\text{updated}}} \leftarrow \pi_\theta$

6. Experiments

6.1. Implementation Details

6.1.1. Training details

Following recent works (Zheng et al., 2025a; Cheng et al., 2025; Zheng et al., 2025c), We employ Qwen2.5-3B (Qwen et al., 2025) and Qwen3-4B-Base (Yang et al., 2025) as our backbone models. Following established protocols (Yu et al., 2025; Cheng et al., 2025; Cui et al., 2025), we utilize the DAPO-Math-17K dataset (Yu et al., 2025) for training. Given its demonstrated stability and superiority over vanilla GRPO (Yu et al., 2025; Cheng et al., 2025), we adopt the DAPO algorithm (Yu et al., 2025) as both our baseline and the core optimization method for our SDRL approach.

We configure the learning rate at 1×10^{-6} with a linear warm-up over the first 10 rollout steps. The rollout phase utilizes a prompt batch size of 256, generating 8 responses per prompt. For each debate pair, we generate 4 responses for Qwen2.5-3B and 8 responses for Qwen3-4B-Base. We utilize a sparse reward signal, assigning $+1$ for correct solutions and -1 otherwise. All training experiments are implemented within the verl framework (Sheng et al., 2024). Additional configuration details are provided in the Appendix.

6.1.2. Evaluation

Benchmarks. Following prior work (Choi et al., 2025; Zhao et al., 2025a; Li & Goyal, 2025), we conduct evaluations on the mathematical reasoning benchmarks MATH500 (Hendrycks et al., 2021), AMC 2023, and AIME 2024/2025.

Multi-Agent Debate. Consistent with established MAD literature (Choi et al., 2025), we primarily evaluate under the *Decentralized MAD* setting (Du et al., 2023; Choi et al., 2025), where each agent observes all peer responses from the previous round. Additional ablation studies on more debate frameworks are provided in Section 6.4. We benchmark performance against the strong *Majority Voting* baseline, which selects the most common answer from the initial round without debate. Our main experiments utilize $N = 5$ agents and $T = 1$ debate round; ablations on N and T are provided separately. Inference is conducted with a rollout temperature of 1.0 and top-p sampling ($p = 0.9$). For all MAD experiments, we report the average score across 5 independent runs.

Table 1. Multi-agent debate performance comparison of different model architectures between the DAPO baseline and SDRL across math reasoning benchmarks. The debate system contains 5 agents. *Maj* denotes the majority-vote accuracy of the agents' direct responses to the question. *Debate* denotes the performance of the decentralized multi-agent system after debate round 1. Δ is the difference between *Maj* and *Debate*. Best results are bolded.

Method	MATH500			AMC23			AIME24			AIME25			Avg.		
	Maj	Debate	Δ	Maj	Debate	Δ	Maj	Debate	Δ	Maj	Debate	Δ	Maj	Debate	Δ
Qwen2.5-3B															
DAPO	70.9	71.1	0.2	49.0	50.0	1.0	7.3	10.0	2.7	4.0	3.3	-0.7	32.8	33.6	0.8
+ SDRL-rand	70.7	71.4	**0.7**	49.2	**53.8**	**4.6**	8.7	11.7	3.0	**5.6**	**6.1**	0.5	33.6	**35.8**	**2.2**
+ SDRL-freq	**71.4**	**71.7**	0.3	**52.5**	53.5	1.0	**9.3**	**13.3**	**4.0**	3.3	4.7	**1.4**	**34.1**	**35.8**	1.7
Qwen3-4B-Base															
DAPO	86.9	83.1	-3.8	**73.0**	76.0	3.0	26.7	28.3	1.6	24.7	24.0	-0.7	52.8	52.9	0.1
+ SDRL-rand	**88.8**	**86.6**	-2.2	72.5	**80.0**	**7.5**	**30.8**	31.7	0.9	22.7	24.0	**1.3**	53.7	55.6	1.9
+ SDRL-freq	87.0	85.9	**-1.1**	72.5	79.0	6.5	28.7	**36.0**	**7.3**	**28.7**	**30.0**	**1.3**	**54.2**	**57.7**	**3.5**

Table 2. Single-agent performance comparison of different model architectures between the DAPO baseline and SDRL across several math reasoning benchmarks. For each benchmark, we report the average accuracy $\text{mean@}K$ and the majority-vote accuracy $\text{maj@}K$ over K independent runs. Best results are bolded.

Method	MATH500		AMC23		AIME24		AIME25		Avg.	
	mean@4	maj@4	mean@32	maj@32	mean@32	maj@32	mean@32	maj@32	mean@32	maj@32
Qwen2.5-3B										
DAPO	64.5	70.4	48.5	56.8	7.2	13.5	3.1	4.2	30.8	36.2
+ SDRL-rand	65.1	70.6	48.1	57.4	**9.9**	13.9	**4.8**	**8.0**	32.0	**37.5**
+ SDRL-freq	**65.5**	**71.0**	**49.9**	**58.0**	9.7	**15.5**	4.1	4.9	**32.3**	37.4
Qwen3-4B-Base										
DAPO	82.9	84.4	71.9	81.4	24.8	31.1	24.9	27.8	51.1	56.2
+ SDRL-rand	**83.7**	85.8	72.3	**84.1**	25.9	**35.1**	20.4	27.6	50.6	58.2
+ SDRL-freq	82.2	**86.2**	**74.2**	83.0	**26.6**	34.4	**26.6**	**37.2**	**52.4**	**60.2**

Single-Agent Reasoning. Following recent RL reasoning literature (Guo et al., 2025a; Liu et al., 2025c), we also assess model performance without debate. For each prompt, we sample K independent responses, and report the average accuracy ($mean@K$) and majority-vote accuracy ($maj@K$), setting $K = 32$ for the AMC and AIME datasets, and $K = 4$ for the MATH500 benchmarks. Further evaluation details are available in the Appendix.

6.2. Multi-Agent Debate

Table 1 compares SDRL with the DAPO baseline under the decentralized MAD setting with N=5 agents and one debate round. SDRL-rand stands for random pairing, and SDRL-freq denotes frequency-based pairing for debate pair construction. Across both backbones, SDRL yields consistent gains in both the post-debate performance (*Debate*) and the first-round majority vote accuracy (*Maj*).

For Qwen2.5-3B, SDRL-rand provides a strong boost to the post-debate stage, improving the average *Debate* accuracy from 33.6 to 35.8 and increasing the average debate gain Δ from 0.8 to 2.2. The gains are most pronounced on the hardest benchmark AIME25, where *Debate* improves from 3.3 to 6.1. SDRL-freq also yields consistent debate benefits across benchmarks, matching the best average *Debate* accuracy of 35.8 while achieving a strong average debate gain of $\Delta = 1.7$. In particular, SDRL-freq substantially improves AIME24, where *Debate* increases from 10.0 to 13.3 and Δ rises from 2.7 to 4.0. For the stronger Qwen3-4B-Base model, DAPO exhibits mixed debate behavior, including degradation on MATH500 and AIME25. SDRL-rand improves overall post-debate performance, raising the average *Debate* accuracy from 52.9 to 55.6 and average *Debate* increases from 0.1 to 1.9. More importantly, SDRL-freq yields the strongest and most robust improvements, raising the average *Debate* accuracy to 57.7 and improving the average debate gain to $\Delta = 3.5$. These gains are driven by challenging benchmarks, most notably on AIME24 where *Debate* increases from 28.3 to 36.0 and Δ rises from 1.6 to 7.3.

Overall, these results demonstrate that SDRL systematically improves decentralized multi-agent debate by making reasoning LLMs better debaters. After training, debate more re-